Department of Electrical and Computer Engineering
Request for Senior Design Project Descriptions

Project title: IoT Based Smart Irrigation and Soil Monitoring System

Project faculty or external advisor: Ridha Kamoua

Description:

The goal of this senior design project is to develop a low-cost, IoT-based system that automates monitoring and management of environmental conditions in small agricultural operations. The system will collect and analyze data such as soil moisture, temperature, humidity, nutrient levels, rainfall, wind, and light conditions to optimize plant growth while reducing water and energy consumption.

To minimize energy usage, the system should be powered by solar energy. Because reliable internet connectivity may not always be available in agricultural environments, the project should also evaluate and implement an appropriate low-power communication protocol that supports reliable data transmission with minimal energy consumption.

Project requirements:

The primary objective is to design and develop a **smart irrigation system** that is:

- Low cost
- Easy to install, configure, operate, and maintain
- Modular and scalable
- Capable of automatically controlling irrigation based on environmental conditions
- Capable of collecting and displaying environmental data
- Equipped with a manual-operation option that allows the user to control irrigation directly

Technical requirements:

Students working on this project should have, or be willing to develop, knowledge and skills in:

- Embedded systems and microcontrollers
- Sensors and data acquisition
- Wireless and low-power communication
- Automation and control systems
- IoT system design
- Artificial intelligence and machine learning
- Software development for dashboards and mobile applications

Resources:

Available Budget: $690

Deliverables and meetings:

Project meetings will be held regularly on **Wednesdays**.
At the conclusion of the project, the following deliverables are expected:

- System architecture and block diagram
- Functional hardware prototype, including environmental sensors and irrigation controls
- User dashboard and mobile application for monitoring data and controlling the system
- Source code for the microcontroller, backend, and associated software

Demonstration of the fully functional system

Recommended team composition (3 students)

Discipline	Recommended Number of Students
Computer Engineering	1–2
Electrical Engineering	1–2
Other / Related Discipline	As appropriate